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**INTERNET PROGRAMMING AND MOBILE
PROGRAMMING**

CEF440

Task 3: Requirement Analysis

GROUP 9

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1. Introduction

This report presents the analysis of requirements gathered for the proposed low-bandwidth e-learning system. The analysis was done using responses collected from 80 participants including students and lecturers. The purpose of this task is to review the gathered requirements, identify conflicts and missing information, classify and prioritise the requirements, then produce a complete Software Requirement Specification (SRS) for the system.

The findings in this report are based on actual survey responses collected from stakeholders and not assumptions.

2. Review and Analysis of Requirements

The collected requirements were reviewed based on completeness, clarity, consistency, feasibility and dependency between requirements.

2.1 Completeness of Requirements

Most respondents clearly identified the major problems they face while using online learning platforms. The responses covered:

- Internet connectivity issues
- High video data consumption
- Missed live classes
- Need for offline learning support
- Need for notifications and reminders
- Desire for downloadable lecture materials

From the survey:

- **81.3%** of users reported missing classes because of poor internet connection.
- **70.3%** requested automatic video quality reduction during weak network conditions.
- Many users also requested downloadable lecture notes and recorded classes.
- These responses provided enough information to define the core system functionalities.

2.2 Clarity of Requirements

Most requirements were understandable, although some responses were too general. For example:

“The app should work better with bad network”

“Videos should not consume much data”

Such responses needed further interpretation before they could be converted into technical requirements.

Some stakeholders also requested “fast access” without specifying acceptable response times. Because of this, measurable performance requirements had to be added during analysis.

2.3 Feasibility Analysis

The proposed requirements were analysed technically and economically.

The following features were considered feasible:

- Adaptive video quality
- Offline content download
- Push notifications
- Recorded lecture storage
- Low-data mode

However, some advanced suggestions like full AI tutoring and automatic real-time translation were considered outside the current project scope because of implementation complexity and limited resources.

2.4 Dependency Between Requirements

Certain requirements depend on others before implementation.

Examples:

Offline learning depends on downloadable content support.

Adaptive streaming depends on network monitoring functionality.

Notifications depend on user account management.

These dependencies were considered during prioritisation.

3. Inconsistencies, Ambiguities and Missing Information

During analysis, some inconsistencies and ambiguities were discovered.

3.1 Identified Inconsistencies

Conflict 1: Live Video Preference vs Low Bandwidth Need

Some users strongly preferred live video classes for interaction, while many also complained that live video consumes too much data and fails during weak network conditions.

Resolution

The system will:

Support live classes when bandwidth is stable

Automatically reduce video quality during weak connection

Provide recorded versions for offline viewing

This resolves the conflict while satisfying both stakeholder groups.

Conflict 2: Automatic Switching vs User Permission

Some respondents wanted the system to automatically switch to low-data mode, while others preferred to be asked first before changes are made.

Resolution

The final requirement states:

The system shall notify users before switching modes for the first time.

Users may later enable automatic switching permanently in settings.

3.2 Ambiguities

Some requirements lacked enough detail.

Examples:

“Fast loading”

“Affordable internet usage”

“Easy navigation”

These were converted into measurable requirements inside the SRS.

Example:

“Fast loading” became:

“The system shall load dashboard pages within 3 seconds under stable network conditions.”

The survey did not clearly define:

- Exact bandwidth threshold for quality switching
- Maximum offline storage size
- Security requirements for downloaded content

These areas still need stakeholder confirmation before moving to system design.

4. Prioritised Requirements (MoSCoW Method)

In prioritising the requirements, we employed the **MoSCoW** technique. In this technique, each letter stands for a priority level:

- **M - Must Have:** Non-negotiable. The system fails without this. Think of it as the minimum viable product.
- **S -Should Have:** Important but not critical for launch. Can survive without it temporarily.
- **C -Could Have:** Nice to have. Only included if time and resources allow.
- **W -Won't Have (this time):** Agreed to be excluded from this version. Not forever, just not now.

4.1 Must Have Requirements

These are critical for system operation.

Requirement

Justification

User login and authentication

Needed for secure access

Low-bandwidth support

Main problem identified by users

Video quality adjustment

Reduces data consumption

Offline access to materials

Helps students with unstable internet

Notifications and reminders

Prevents missing classes

4.2 Should Have Requirements

Requirement

Justification

Recorded lecture storage

Improves learning flexibility

Discussion forum

Encourages interaction

Download progress indicator

Improves usability

4.3 Could Have Requirements

Requirement

Justification

Dark mode

User comfort

Personal learning statistics

Additional support feature

Multi-language interface

Future scalability

4.4 Won't Have for Current Version

Requirement

Reason

AI tutoring assistant

High implementation complexity

Real-time language translation

Outside current scope

5. Functional and Non-Functional Requirements

5.1 Functional Requirements (what the system must do)

FR-01: User Authentication

The system shall allow users to create accounts and login securely.

FR-02: Adaptive Video Streaming

The system shall automatically reduce video quality during weak internet connection.

FR-03: Offline Access

Users shall be able to download lecture materials and recorded classes for offline use.

FR-04: Notifications

The system shall send reminders for upcoming classes and assignments.

FR-05: Lecture Recording

The system shall store recorded lectures for later viewing.

FR-06: Discussion Forum

The system shall provide communication between students and lecturers.

5.2 Non-Functional Requirements (how well the system does it)

NFR-01: Performance

Dashboard pages should load within 3 seconds under stable internet conditions.

NFR-02: Availability

The system should be available 24/7 except during maintenance periods.

NFR-03: Security

User information and downloaded materials must be protected from unauthorized access.

NFR-04: Usability

The interface should be simple and easy for first-time users to understand.

NFR-05: Reliability

Offline downloads should synchronize properly once internet connection is restored.

6. Software Requirement Specification (SRS)

6.1 System Overview

The proposed system is a low-bandwidth e-learning platform designed to support students and lecturers in areas with unstable internet connectivity. The platform focuses on reducing data usage while maintaining access to educational resources.

6.2 Stakeholders

The main stakeholders include:

- Students
- Lecturers
- System administrators
- University management

6.3 System Features

- User Management
- Account registration
- Login and logout
- Password recovery
- Learning Features
- Live classes
- Recorded lectures
- Downloadable materials
- Assignment submission
- Bandwidth Optimisation
- Automatic video quality adjustment
- Low-data mode
- Offline access support
- Communication Features
- Notifications
- Discussion forums
- Lecturer announcements

6.4 Constraints

- Limited internet infrastructure in some regions

- Limited storage space on student devices
- Financial constraints affecting mobile data access

7. Stakeholder Validation

The analysed requirements were reviewed against stakeholder responses collected from the survey. Most identified needs were properly represented in the final requirement list.

However, the following still require confirmation from stakeholders:

- Security expectations for downloaded content
- Offline synchronisation behaviour
- Exact bandwidth threshold values for adaptive streaming

8. Conclusion

This task successfully analysed and organised the gathered system requirements into a structured Software Requirement Specification. The report identified major stakeholder concerns, resolved requirement conflicts and prioritised system functionalities according to project importance.

The findings show that low-bandwidth optimisation, offline access and adaptive streaming are the most critical features required by users. These requirements will guide the next phase of system design and implementation.